Discrete Mathematics

Unit 5:
Relation and graphs

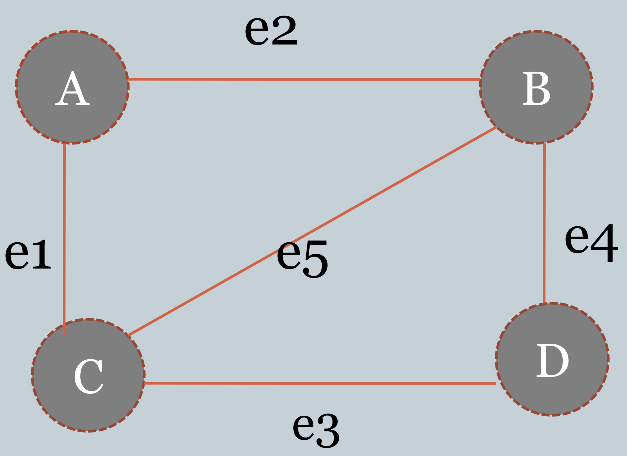
5.2 GRAPH THEORY

- **Graph:** A graph G= (V,E) consists of a V, a nonempty set of vertices(nodes) and E, a set of edges. An edge is connected to its vertices.

G=(V, E)

V={A, B, C, D}

E={e1, e2, e3, e4, e5 }



Application of Graph

Graphs are used to solve problems in many fields. Some of them are as follows:

- Graphs are used to model the geographic maps of cities in which each place in city can be represented by the node and the road connecting such places are represented by an arc (edges)
- Graphs are used to model the computer network in which each node is a machine (computer, hub, router, switch etc) and the link between them represents the edge.
- It can help to solve typical problems those concerned with finding shortest path or most economical route between two vertices, or the smallest set of edges which connect all the vertices in a graph.
- It is used to find the number of different combination of flight between two cities in an airline network.

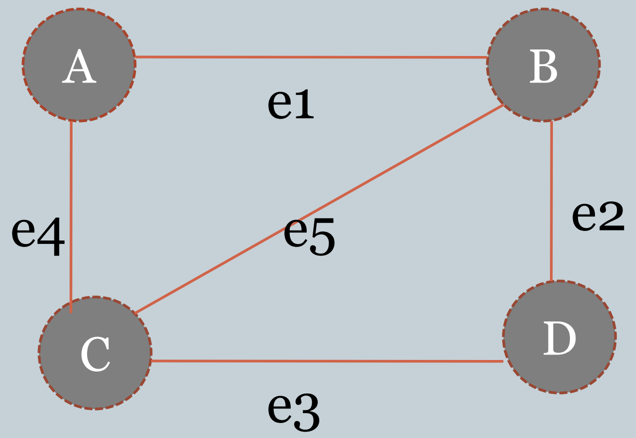
- **Infinite Graph and Finite Graph:**

  A graph with an infinite vertex set is called an infinite graph and a graph with a finite vertex set is called a finite graph

- **Simple Graph:** A graph in which each edge connects two different vertices and where no two edges connect the same pair of vertices is called simple graph.

i.e. A graph G=(V,E) with no loops and no parallel edges.

Simple graph cont..



V(G)= { A,B,C,D}

E(G)= {e1, e2, e3, e4, e5}

e1= { A,B}, e2={ B,D}, e3={ C,D}, e4={ A,C}

e5={ B,C}

Here, none of the edges are in loop and doesn't have parallel edge.

Hence, it is simple graph.

d(A)+d(B)+d(C)+d(D)=2|E|

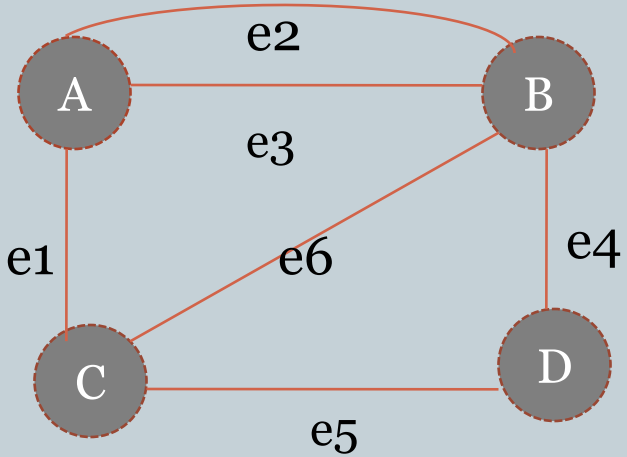
3+3+3+3=2*6

12=12

Multi Graph

Multi Graph:

A graph that may have multiple edges connecting the same vertices are called multi graph.



V(G)={ A,B,C,D}

E(G)={e1,e2,e3,e4,e5,e6}

e1={ a, c}, e2={a, b}, e3={a, b}

The edge e2 and e3 have same vertices so it is

Multi graph.

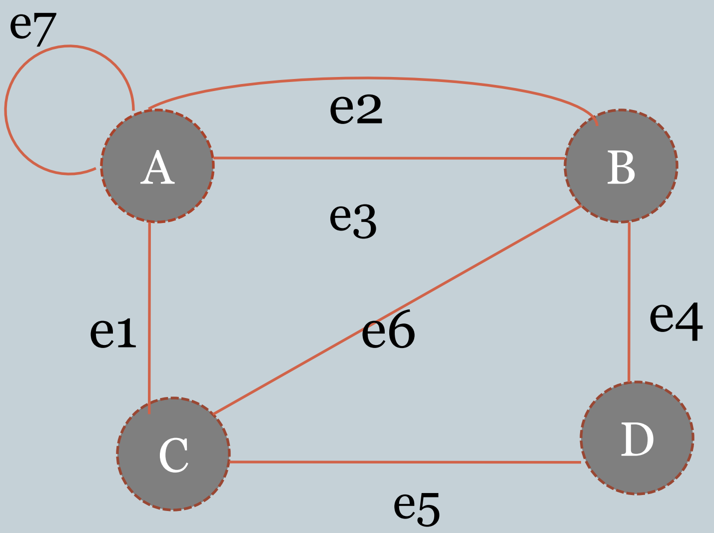
Pseudograph :

Graph that may include loops and possibly multiple edges connecting the same pair of vertices are called pseudo graph.

e7={a, a}

e2={a, b}

e3={a, b}

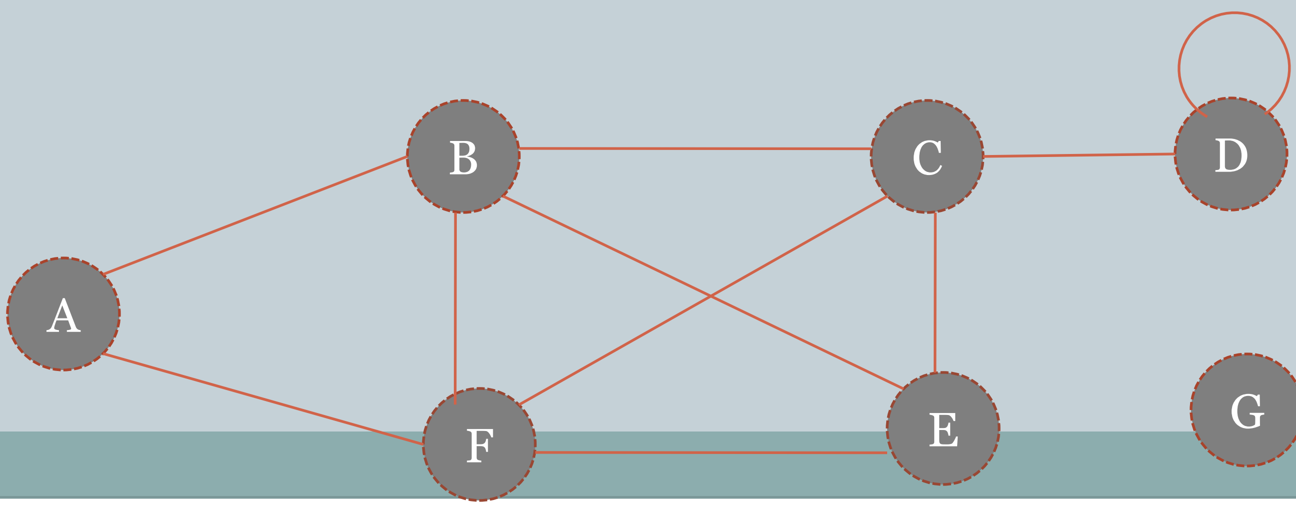


Graph Terminologies

Order and size of graph: If G=(V,E) be a finite graph then the no. of vertices in graph is called order of graph and the no. of edges in graph is called size of graph.

Degree of vertex: The degree of vertex in an undirected graph is the number of edges incident with it, except that a loop at a vertex contributes twice to the degree of that vertex.

It is denoted by deg().



Here, the degree of the vertices

deg(a)= 2, deg(b)=4, deg(c)= 4, deg(d)= 1+2=3

deg(e)= 3, deg(f)=4, deg(g)= 0

Isolated vertex: A vertex of degree zero is called isolated vertex.

Pendant vertex: A vertex having degree one is called pendant vertex.

Adjacent vertices & edges:

- Two vertices U & V are called adjacent if there is an edge between U & V.
- Two edges e1 & e2 are called adjacent if they share a common vertex.

Types of graph:

Trivial graph:

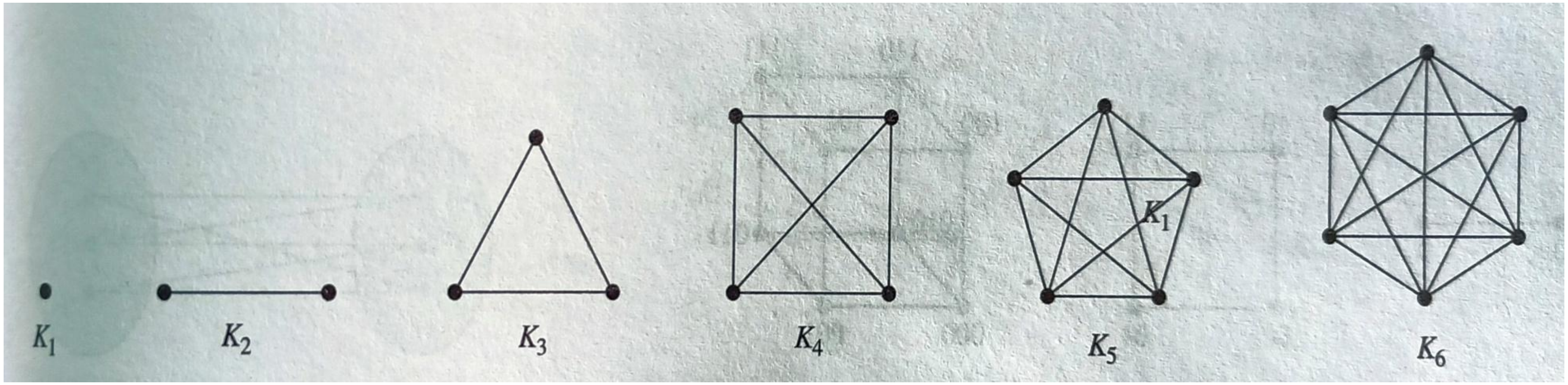
A graph with one vertex & no edge is called a trivial graph.

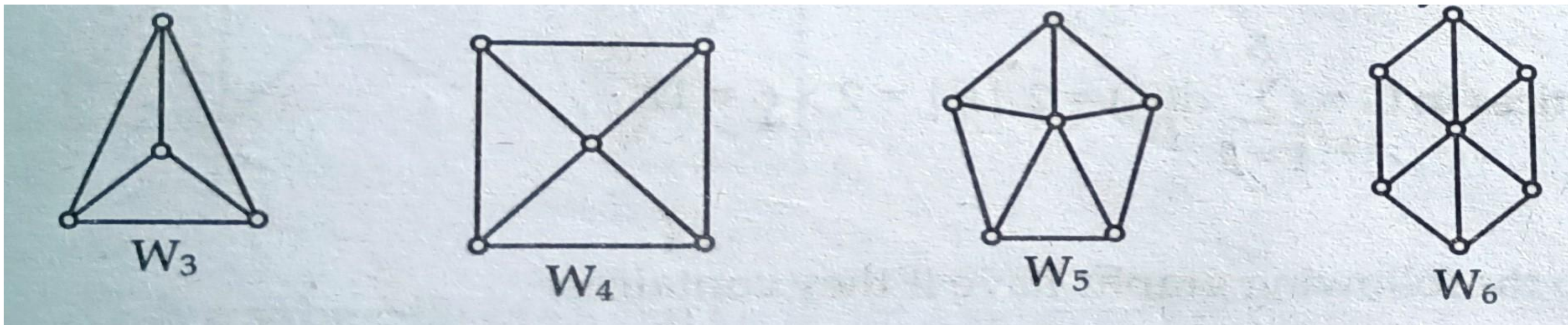
Complete graph:

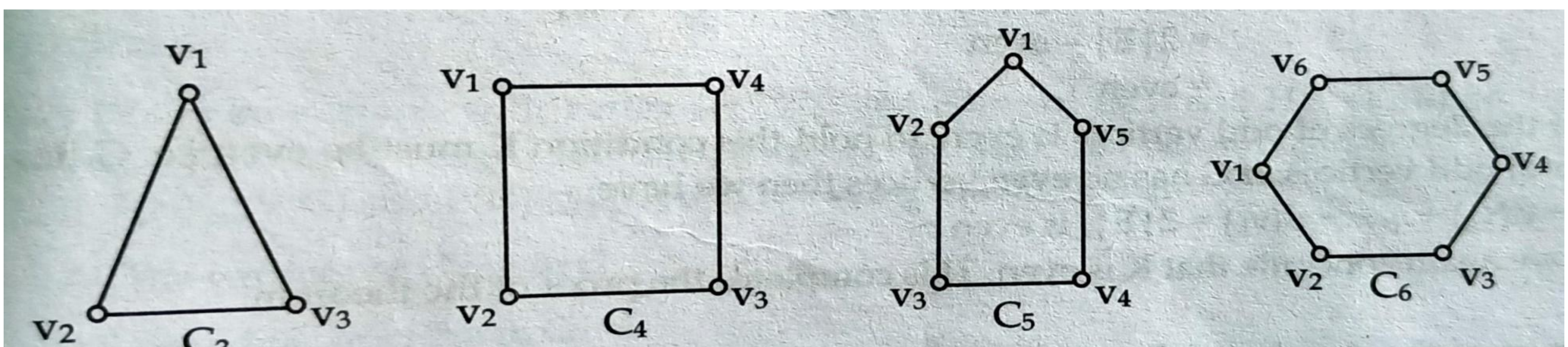
A graph G is said to be complete if it contains exactly one edge between each pair of distinct vertices. The complete graph with ‘n’ vertices is denoted by K_n

Cycle graph:

A graph with $n \geq 3$ (vertex) where every vertex is connected with other two vertices on either side and last vertex is connected with remaining one(first one) to form a closed path is called cycle graph. The cycle graph with ‘n’ vertices is denoted by C_n .


K_1
K_2
K_3
K_4
K_5
K_6


W_3
W_4
W_5
W_6


v_1 v_2 v_3
C_3
v_1 v_2 v_3 v_4
C_4
v_1 v_2 v_3 v_4 v_5
C_5
v_1 v_2 v_3 v_4 v_5 v_6
C_6

 # Wheel graph:

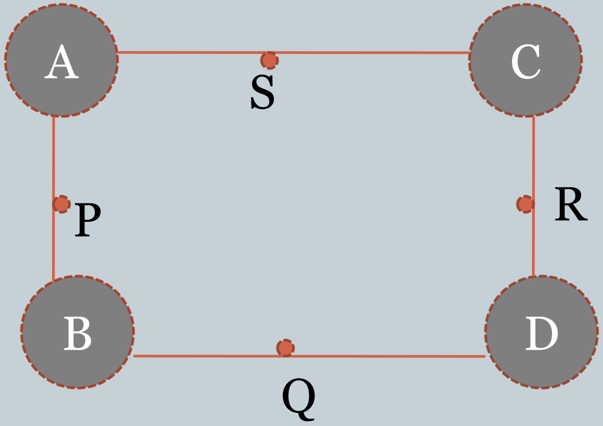
It is formed by an additional vertex to the cycle graph where the new vertex is connected by each vertex of the cycle. It is denoted by W_n

 # Regular graph: A graph in which all vertices have equal degree is called a regular graph. If the degree of each vertex is 'r' then the graph is a r- regular graph.

#Bipartite graph: A simple graph G is said to be bipartite if its vertex set V can be partitioned into two disjoint sets V1 and V2 such that each edge of G connect the vertex in V1 and a vertex in V2 so that no edge in G connect two vertices in V1 or two vertices in V2. when this condition holds we call the pair (V1, V2) a bipartition of the vertex set V of G

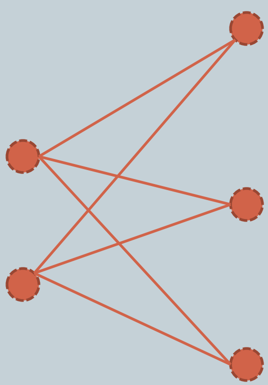
Bipartite sets are {a, b, c, d}

{p, q, r, s}

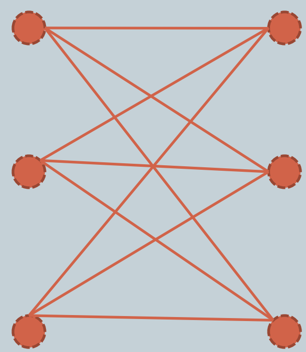


 # Complete Bipartite graph:

- A bipartite graph G is said to be complete if each vertex of first bipartite set is connected to every vertices of another bipartite set.
- A complete bipartite graph with ‘x’ number of vertices in first bipartite set and ‘y’ number of vertices in second bipartite set then it is denoted by $K_{x,y}$.



$K_{2,\ 3}$



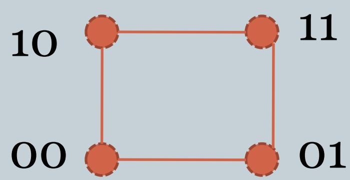
$K_{x,y}$ =?

n- cube graph: The n- cube denoted by Q_n is the graph that has vertices representing the 2^n bits strings of length n .

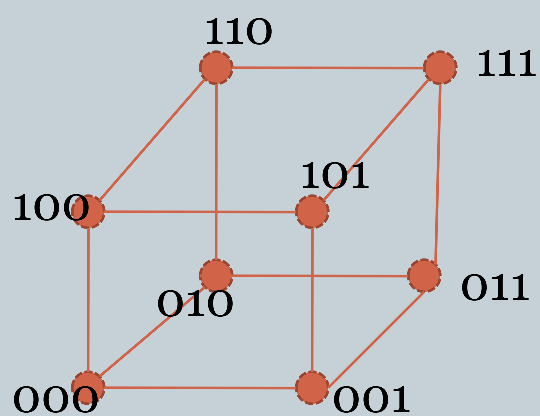
- Two vertices are adjacent iff the bit strings that they represent differ in exactly one bit position.

Eg: for 1 bit string :

For 2 bit string:



For 3 bit string:



Theorem: “The Handshaking”

“The sum of the degree of the vertices of a graph is equal to twice the number of edges”

Proof:

Consider a graph G=(V, E) with n vertices v_1 v_2 v_3………v_n and |E|, the no. of edges in G.

Since, each edge contributes a degree of 2, the sum of the degree of all the vertices in G is twice the no. of edge in G.

i.e.

$$\sum_{i=1}^{n} d(vi) = 2|E|$$

“The no. of odd vertices in a graph is always even”

Proof:

let G be a graph. If G contain no odd(even) vertices, then the result follows immediately.

Suppose G contains ‘K’ number of odd vertices then the vertices are $v_1\ v_2\ v_3 \ldots\ldots\ldots v_k$.

Similarly, if it contains ‘n’ number of even vertices then the vertices $u_1\ u_2\ u_3 \ldots\ldots\ldots u_n$.

So, $d(u_1) + d(u_2) + \ldots\ldots\ldots + d(u_n) =$ even

By handshaking theorem we can conclude that

$$\sum_{i=1}^{n} d(vi) = 2|E|$$

where E is the total no of edge in graph G.

or, $[\, d(v_1) + d(v_2) + \ldots\ldots\ldots + d(v_k)\,] + [\, d(u_1) + d(u_2) + \ldots\ldots\ldots + d(u_n)\,] = 2\,|E|$

or, $d(v_1) + d(v_2) + \ldots\ldots\ldots + d(v_k) = 2\,|E|$ - Even

= Even

Here, sum of the degree of odd vertices is even, to hold this condition ‘K’ must be even i.e. G has an even number of odd vertices.

If G has no even(odd) vertices then we have

$$d(v_1) + d(v_2) + \ldots\ldots\ldots + d(v_k) = 2\,|E|$$

which is even

From which we again conclude that ‘K’ is even.

Hence, the no. of odd vertices in a graph is always even . Proved.

How many vertices do the following graphs have?

a) 16 edges and all vertices of degree 2.

b) 21 edges, 3 vertices of degree 4 and all other vertices of degree 3.

C) 20 edges, 9 vertices of degree 2, and other vertices of degree 5.

d) 25 edges and 2 vertices of degree 5, 3 vertices of degree 4 and other vertices of degree 2.

a) soln:

No of edge(E)= 16

Let there are ‘n’ no. of vertices each of degree 2

Now,

from handshaking theorem

$\deg(v_i) = 2|E|$

or, $n * 2 = 2 * 16$

or, $n = 16$

Therefore 16 vertices.

b) soln:

No of edge(E)= 21

Let there are ‘n’ no. of vertices

Since, 3 vertices have degree of 4

Hence, remaining vertices i.e (n-3) must have degree 3.

Now,

from handshaking theorem

$\deg(v_i) = 2|E|$

or, $\deg(3) + \deg(n-3) = 2|E|$

or, $3 * 4 + (n-3) * 3 = 2*21$

or, $12 + 3n - 9 = 42$

or, $3n = 39$

or, $n = 13$

Therefore 13 vertices.

The total number of edges in a complete graph K_n is $n(n-1)/2$.

proof:

From handshaking theorem ,

$$\Sigma \deg(v_i) = 2|E|$$

Where |E| is the number of edges in G with n vertices.

or, $d(v_1) + d(v_2) + d(v_3) + \ldots\ldots + d(v_n) = 2|E|$

Since maximum degree of each vertex in the graph G can be (n-1)

Then,

$(n-1) + (n-1) + \ldots\ldots$ to n terms $= 2|E|$

or, $n(n-1) = 2|E|$

$$E = n(n-1)/2$$

Hence proved,

The order and size of the complete bipartite graph $K_{m,n}$ is m+n and mn respectively.

Proof:

The complete bipartite graph $K_{m,n}$ is formed from two partition sets A and B having 'm' and 'n' number of vertices respectively.

So, number of vertices in $K_{m,n}$ is the sum of vertices in A and B respectively,

Therefore order of $K_{m,n}$ = m+n.

Again, in complete bipartite graph $K_{m,n}$ each vertex of partition set A is connected to each vertex of partition set B, so, there are 'n' vertices in set B each having degree 'm' thus there will be n*m sum of degree of vertices in set B.

Similarly, in set A there are 'm' vertices each with degree 'n' so degree of vertices in set A is m*n.

Now, From handshaking theorem ,

$\Sigma \deg(v_i) = 2|E|$ Where |E| is the number of edges in G.

Or, $d(A) + d(B) = 2|E|$

Or, $(m*n) + (n*m) = 2|E|$

Or, $2(m*n) = 2|E|$

$m*n = |E|$

Therefore size of $K_{m,n}$ = m*n.

Subgraph:

Let G=(V,E) and H= (V,E) be two graphs then H is said to be subgraph of G if $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$.

It is written as $H \subseteq G$.

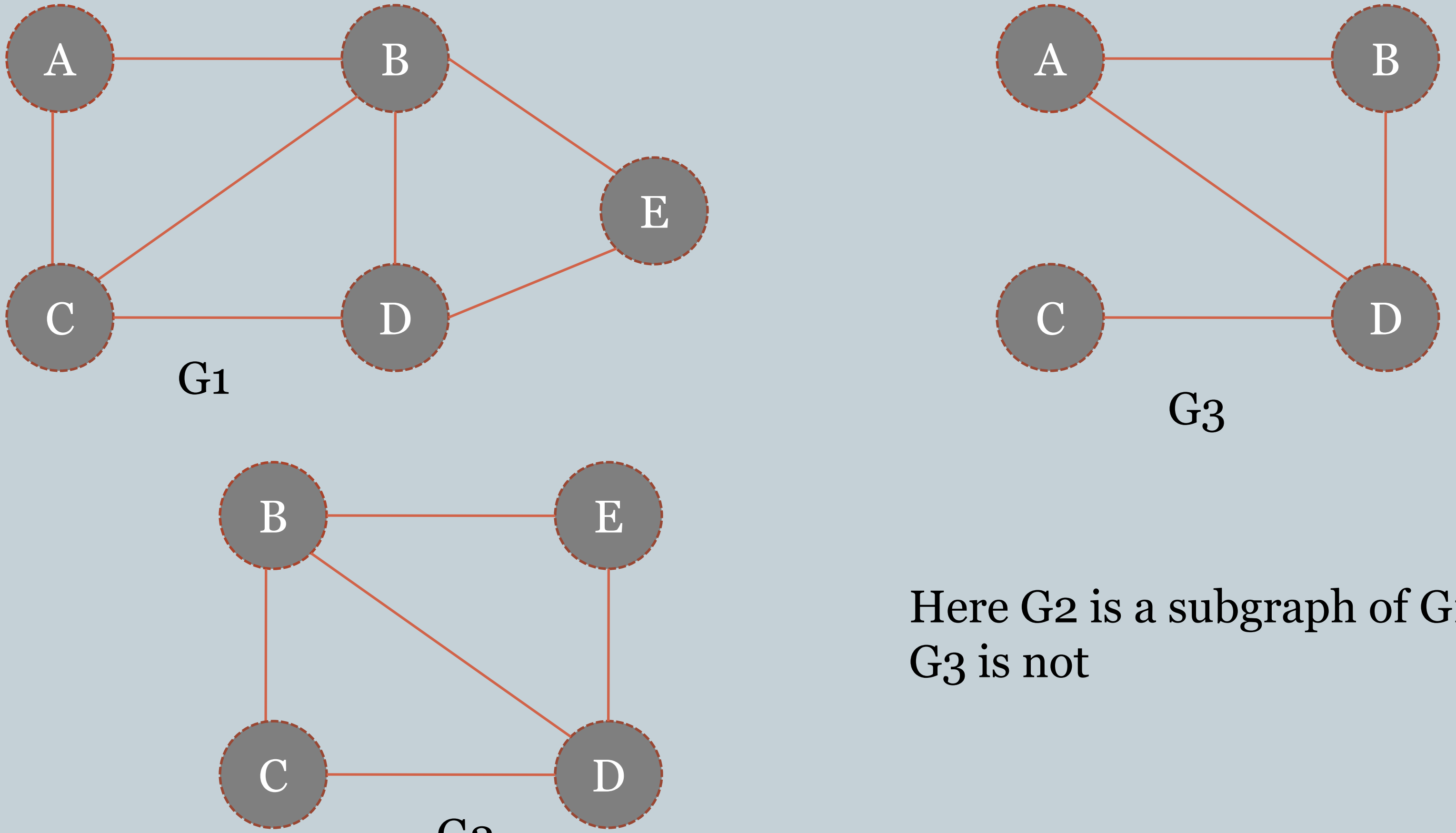
If H is a subgraph of G then,

- All edges of H are in G.
- All vertices of H are in G.
- Each edge of H has the same end points in H as in G.

#Spanning subgraph: The subgraph H of graph G is said to be spanning subgraph if $V(H) = V(G)$

i.e. H and G has exactly same vertex set.

subgraph



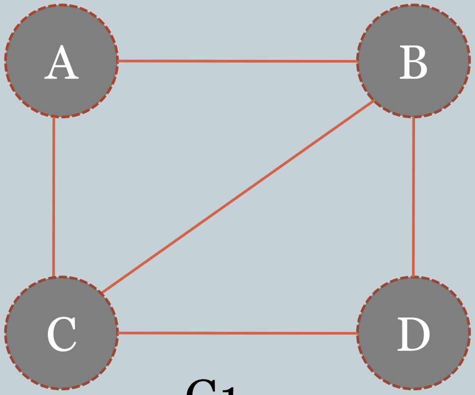
G1

G3

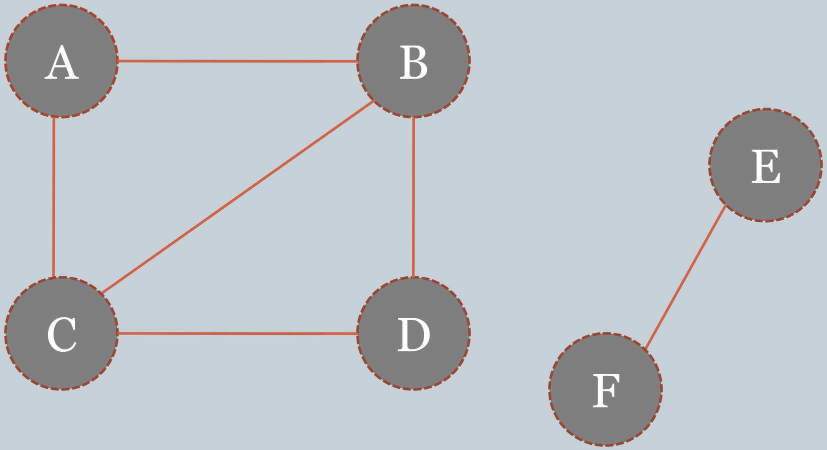
G2

Here G2 is a subgraph of G1 but G3 is not

#Connected graph: A graph is said to be connected if there is a path between each possible pair of its vertices otherwise G is disconnected.



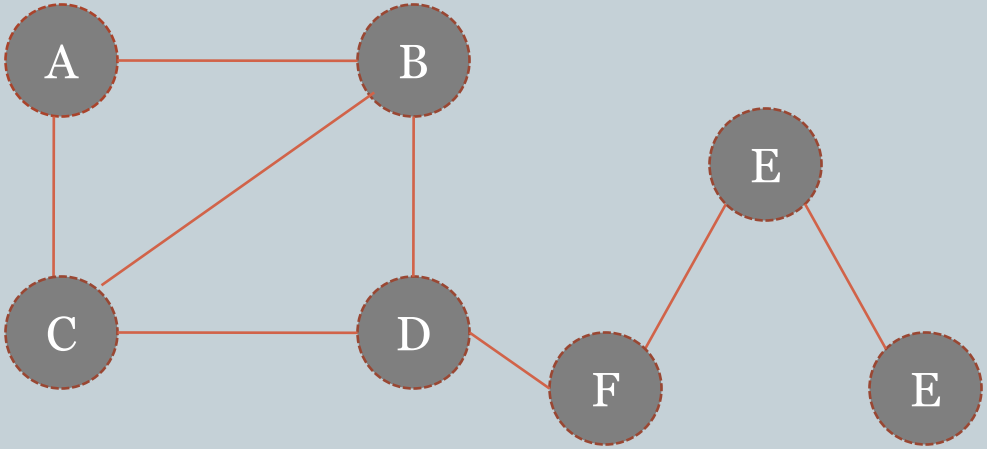
G1



G2

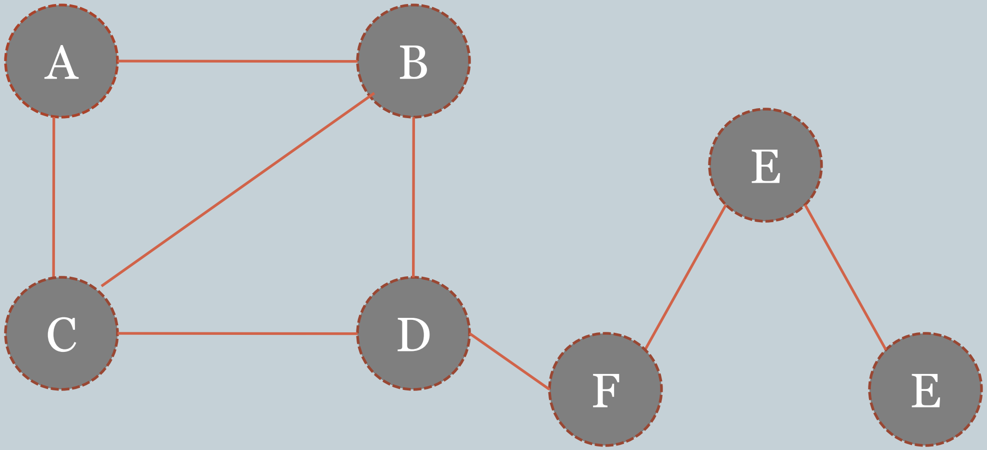
Here G1 is connected Graph and G2 is disconnected graph.

#Cut-vertex: A vertex in a graph is said to be cut vertex if graph becomes disconnected after removal of this vertex from the graph.



Here the vertex f is cut vertex since removal of it a graph G become disconnected.

#Cut edge or bridge: A edge in a graph after removal graph becomes disconnected then such edge is called cut edge.



Here the edge (d, f) is cut edge since removal of it a graph G become disconnected.

Representation of graph

Graph can be represented in matrix based on adjacency of vertices and incidence of vertices and edge.

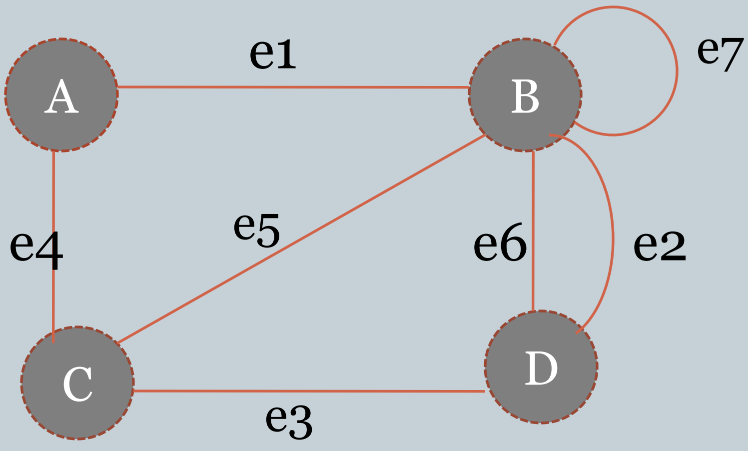
If graph is represented by adjacency of vertices then it is called **adjacency matrix** and if it is represented by incidence of vertices and edges then it is called **incidence matrix**.

#Adjacency Matrix:

Let G=(V, E) be a graph with ‘n’ vertices v_1 v_2 v_3………v_n . The adjacency matrix of G w.r.t given ordered list of vertices is a n*n matrix denoted by A(G)= $(a_{ij})_{n*n}$ such that

$$a_{ij} = \begin{cases} 0 & \text{if there is no edge between vertices.} \\ 1 & \text{if there is an edge between vertices.} \\ K & \text{if there are} \geq 2 \text{ edges between vertices.} \end{cases}$$

eg: Find the adjacency matrix of given graph.



$$A(G) = \begin{array}{c} \\ A \\ B \\ C \\ D \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & D \end{array} \\ \begin{Bmatrix} 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 2 \\ 1 & 1 & 0 & 1 \\ 0 & 2 & 1 & 0 \end{Bmatrix} \end{array}$$

Since the graph has four vertices the Adjacency matrix will be 4*4 matrix.

#Incidence Matrix: let G=(V,E) be an undirected graph. Suppose that $v_1\ v_2\ v_3 \ldots\ldots\ldots v_m$ are the vertices and $e_1\ e_2\ e_3 \ldots\ldots\ldots e_n$ are the edges of G. Then the incidence matrix w.r.t the ordering of V & E is the m*n matrix with $I(G) = (m_{ij})_{m*n}$

Where

$$m_{ij} = \begin{cases} 1 \text{ if } e_j \text{ is incident with } v_i \\ 0 \text{ if } e_j \text{ is not incident with } v_i \\ 2 \text{ if } v_i \text{ is the end of the loop } e_j \end{cases}$$

The G has 4 vertices and 7 edges so,

$$I(G) = \begin{array}{c} \\ A \\ B \\ C \\ D \end{array} \begin{array}{c} \begin{array}{ccccccc} e1 & e2 & e3 & e4 & e5 & e6 & e7 \end{array} \\ \left\{ \begin{array}{ccccccc} 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 \end{array} \right\} \end{array}$$

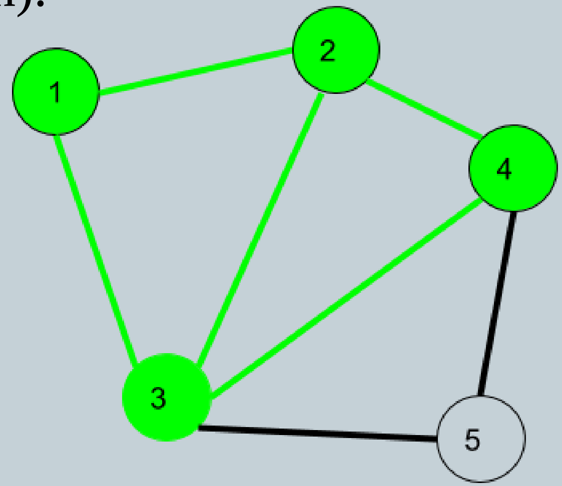
Graph Connectivity

1) Walk:

A walk is a sequence of vertices and edges of a graph i.e. if we traverse a graph then we get a walk. W=(v0, e0, v1, e1,………vn).

Edges can be repeated.

Vertex can be repeated.



Here 1->2->3->4->2->1->3 is a walk

Walk can be open or closed.

Walk can repeat anything (edges or vertices).

a) Open walk:

A walk is said to be an open walk if the starting and ending vertices are different i.e. the origin vertex and terminal vertex are different.

b) Closed walk:

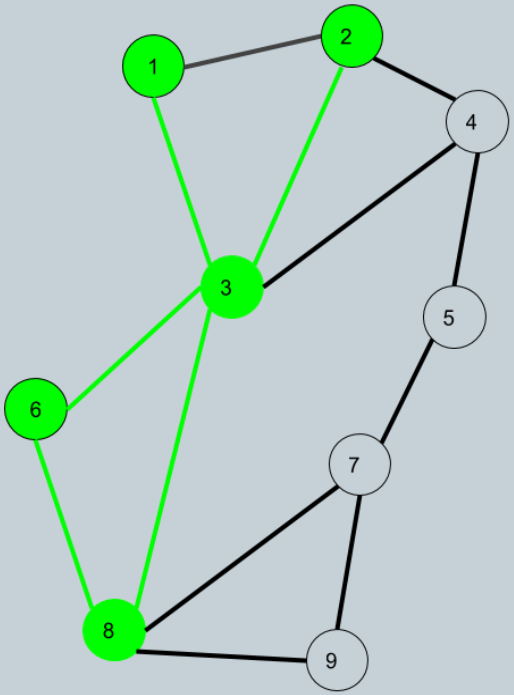
A walk is said to be a closed walk if the starting and ending vertices are identical i.e. if a walk starts and ends at the same vertex, then it is said to be a closed walk.

In the above diagram:
1->2->3->4->5->3-> is an open walk.
1->2->3->4->5->3->1-> is a closed walk.

2) Trail :

Trail is an walk in which no edge is repeated.

Vertex can be repeated.



Here 1->3->8->6->3->2 is open trail.

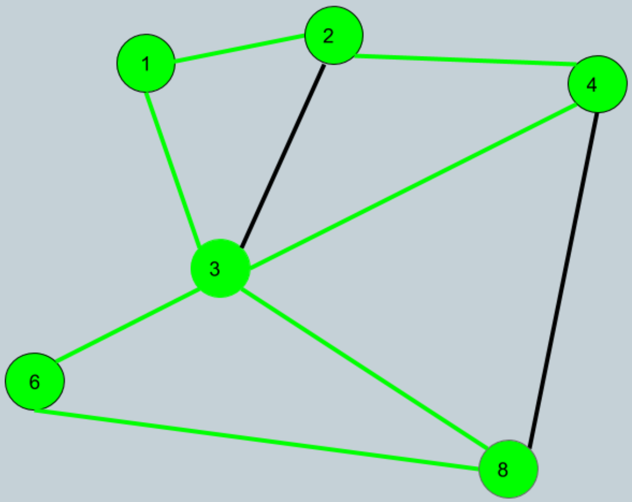
Also 1->3->8->6->3->2->1 will be a closed trail.

3. Circuit :

Traversing a graph such that not an edge is repeated but vertex can be repeated and it is closed also i.e. it is a closed trail.

Vertex can be repeated.

Edge can not repeated.



Here 1->2->4->3->6->8->3->1 is a circuit.

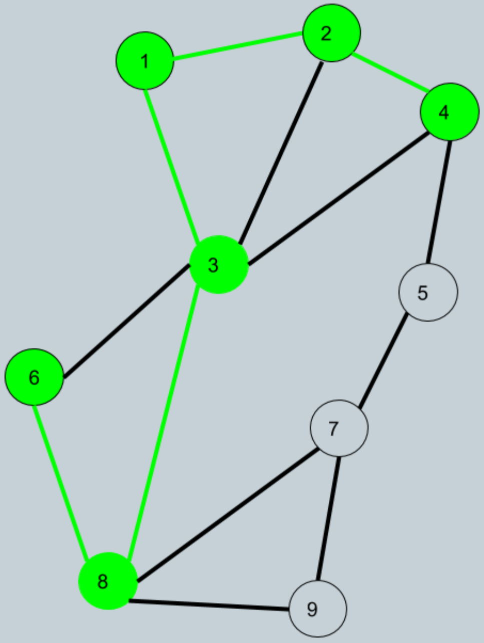
Circuit is a closed trail. These can have repeated vertices only.

4) Path:

It is a walk in which neither vertices nor edges are repeated i.e. if we traverse a graph such that we neither repeat a vertex nor we repeat an edge.

Vertex can not be repeated.

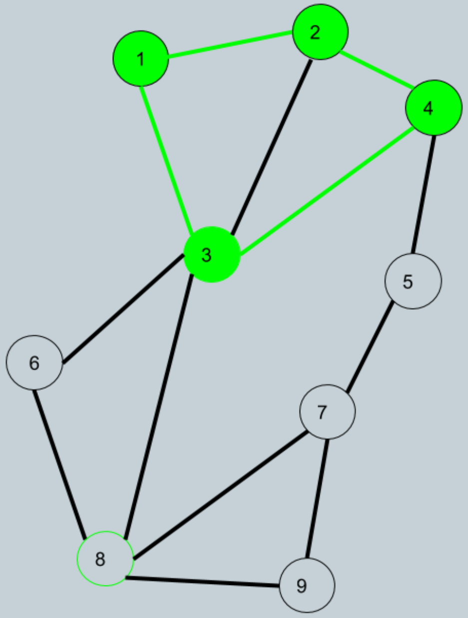
Edge can not be repeated.



Here 6->8->3->1->2->4 is a Path.

5) Cycle:

A circuit which doesn’t repeat any vertices(except initial and final vertex) is called a cycle.
i.e. Traversing a graph such that we do not repeat a vertex nor we repeat a edge
but the starting and ending vertex must be same.
Vertex can not be repeated.
Edge can not be repeated.



Here 1->2->4->3->1 is a cycle.
Cycle is a closed path.
These can not have repeat anything (neither edges nor vertices).

Term	Repeated Edge	Repeated Vertex
Path	No	No
Trail	No	Yes
Walk	Allowed	Allowed
Circuit	No	Yes
Cycle	No	First & Last only

Isomorphism of Graph

- Two graph are said to be isomorphic if they perhaps the same graphs just drawn differently with different names i.e. they have identical behavior for any graph theoretic.
- There is one to one correspondence between their vertices and edges and incidence relationship is preserved.
- G1(V1, E1) and G2(V2, E2) are isomorphic if there exists a bijective mapping ϕ between V1(G1) and V2(G2) such that $\{u, v\}$ is in E1 iff $\{\phi(u), \phi(v)\}$ is in E2. The function ϕ is called isomorphism.
- If two graph G1 and G2 are isomorphic then they must have:

  a) same number of vertices. i.e. $|V1| = |V2|$

  b) same number of edges i.e. $|E1| = |E2|$

  c) if $\{u, v\}$ is in E1 then $\{\phi(u), \phi(v)\}$ is in E2.

  d) if $\deg(u) = K$ in G1 then $\deg(\phi(u)) = K$ in G2.

  e) $A(G1) = A(G2)$ w.r.t orders of vertices $v1, v2, \ldots\ldots vn$ and $\phi(v1), \phi(v2), \ldots\ldots \phi(vn)$

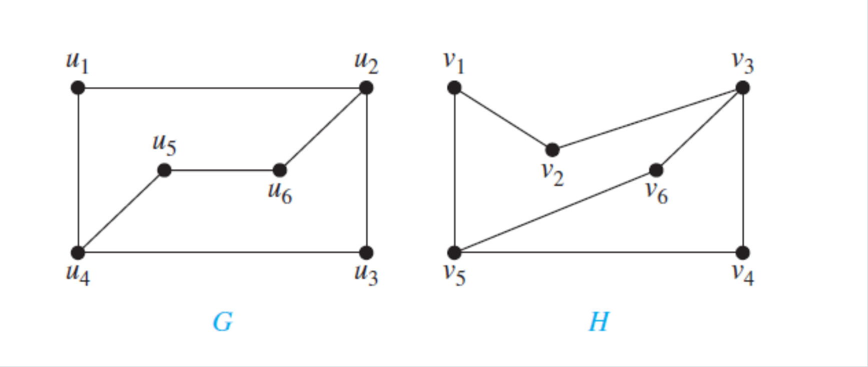
Isomorphism of Graph

Determine whether the graphs *G and H are isomorphic*.

Soln:

No of vertices in G= V(G)= 6

No of vertices in H= V(H)=6



No of edge in G= E(G)= 7

No of edge in H=E(H)= 7

Because G and H agree with respect to these invariants, it is reasonable to try to find an isomorphism ϕ .

Now, We now will define a function ϕ and then determine whether it is an isomorphism.

So,

ϕ(u1)=v6
ϕ(u2)=v3
ϕ(u3)=v4
ϕ(u4)=v5
ϕ(u5)=v1
ϕ(u6)=v2

(u1, u2)=e	(ϕ(u1), ϕ (u2))= (v6, v3)= e
(u2, u3)=e	(v3, v4)=e
(u2, u6)=e	(v3,v2)=e
(u3, u4)=e	(v4, v5)=e
(u4, u1)=e	(v5, v6)=e
(u4, u5)=e	(v5, v1)=e
(u5, u6)=e	(v1, v2)=e

Isomorphism of Graph

Now, the degree of each vertices in G and H.

deg(u1)= 2	deg(ϕ(u1))=deg(v6)=2
deg(u2)= 3	deg(v3)= 3
deg(u3)= 2	deg(v4)= 2
deg(u4)= 3	deg(v5)= 3
deg(u5)=2	deg(v1)=2
deg(u6)=2	deg(v2)=2

deg(u1) = 2 and because u1 is not adjacent to any other vertex of degree two, the image of u1 must be either v4 or v6, the only vertices of degree two in H not adjacent to a vertex of degree two. We arbitrarily set ϕ(u1) = v6. [If we found that this choice did not lead to isomorphism, we would then try ϕ(u1) = v4.] Because u2 is adjacent to u1, the possible images of u2 are v3 and v5. We arbitrarily set ϕ(u2) = v3. Continuing in this way, using adjacency of vertices and degrees as a guide, we set ϕ (u3) = v4, ϕ(u4) = v5, ϕ(u5) = v1, and ϕ(u6) = v2.We now have a one-to-one correspondence between the vertex set of G and the vertex set of H, namely, ϕ(u1) = v6, ϕ(u2) = v3, ϕ(u3) = v4, ϕ(u4) = v5, ϕ(u5) = v1, ϕ(u6) = v2. To see whether ϕ preserves edges, we examine the adjacency matrix of G,

Isomorphism of Graph

$$\mathbf{A}_G = \begin{array}{c} \\ u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \end{array} \begin{array}{c} \begin{array}{cccccc} u_1 & u_2 & u_3 & u_4 & u_5 & u_6 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \end{bmatrix} \end{array},$$

and the adjacency matrix of H *with the rows and columns labeled by the images of the corresponding* vertices in G,

$$\mathbf{A}_H = \begin{array}{c} \\ v_6 \\ v_3 \\ v_4 \\ v_5 \\ v_1 \\ v_2 \end{array} \begin{array}{c} \begin{array}{cccccc} v_6 & v_3 & v_4 & v_5 & v_1 & v_2 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \end{bmatrix} \end{array}.$$

Because A(G)= A(H), it follows that ϕ preserves edges. We conclude that ϕ is an isomorphism,

so G and H are isomorphic. Note that if ϕ turned out not to be an isomorphism, we would not have established that G and H are not isomorphic, because another correspondence of the vertices in G and H may be an isomorphism.

Operations of Graph

Union:

Two graph G and H, then their union will be a graph such that

$$V(G \cup H)= V(G) \cup V(H)$$

And $$E(G \cup H)= E(G) \cup E(H)$$

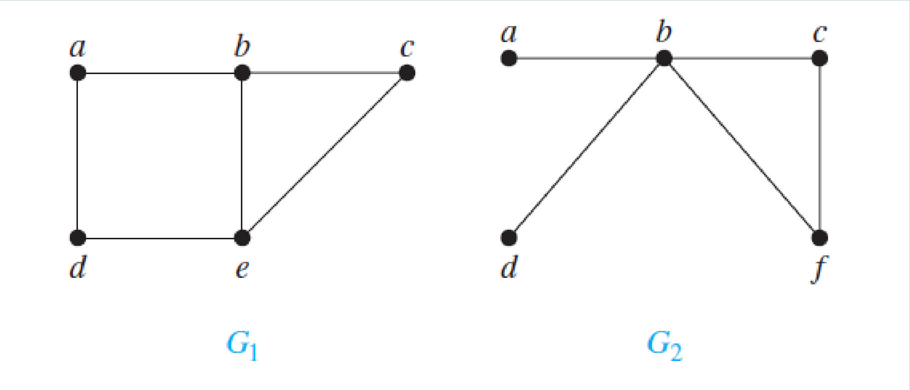
Intersection:

Two graph G and H, then their intersection will be a graph such that

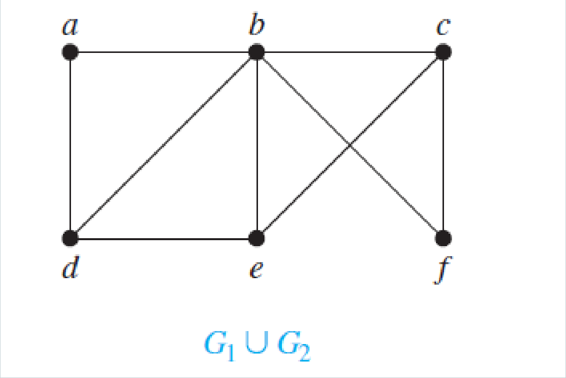
$$V(G \cap H)= V(G) \cap V(H)$$

And $$E(G \cap H)= E(G) \cap E(H)$$

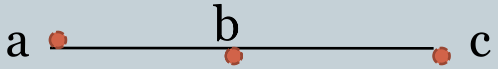
Consider the graphs G1 and G2



Then G1∪ G2 is,

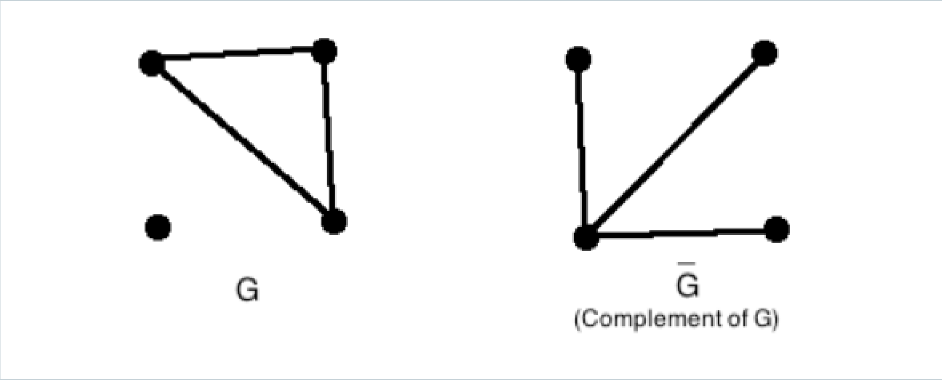


Then G1 ∩ G2 is,



Complement graph:

The **complement** or **inverse** of a graph *G* is a graph *H* on the same vertices such that two distinct vertices of *H* are adjacent iff they are not adjacent in *G*.



Self-complement graph:

A graph is G is self-complement if G is isomorphic to its complement graph.

Directed graph

A directed graph or digraph G=(v, e) consists of two things.

i) A set V=V(G) whose elements are called vertices, nodes of G.

ii) A set E=E(G) of ordered pair (u, v) of vertices called arcs or directed edges.

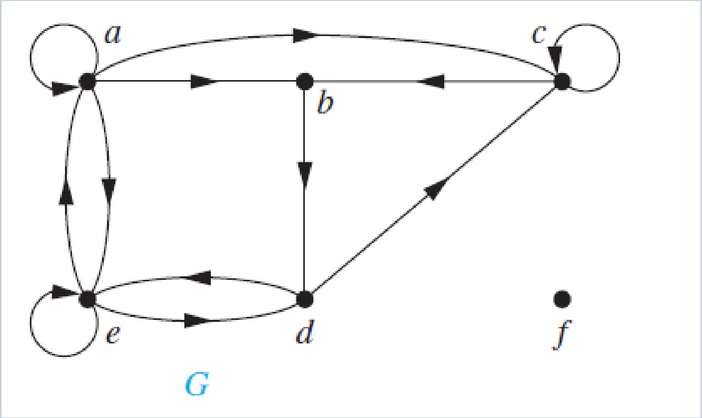
Adjacent vertices:

- When (u, v) is an edge of the graph G with directed edges, u is said to be adjacent to v and v is said to be adjacent from u.
- The vertex u is called the initial vertex of (u, v), and v is called the terminal or end vertex of (u, v).
- The initial vertex and terminal vertex of a loop are the same.

In degree and out degree of vertex:

- In a graph with directed edges the in-degree of a vertex v, denoted by deg−(v) or in-deg(v), is the number of edges with v as their terminal vertex i.e. the number of edges incident towards v.
- The out-degree of v, denoted by deg+(v) or out-deg(v), is the number of edges with v as their initial vertex i.e. the number of edges directed away from v.
- (Note that a loop at a vertex contributes 1 to both the in-degree and the out-degree of this vertex.)

Find the in-degree and out-degree of each vertex in the graph *G* with directed edges.



Solution:

The in-degrees in G are deg−(a) = 2,
deg−(b) = 2,
deg−(c) = 3,
deg−(d) = 2,
deg−(e) = 3,
and deg−(f) = 0.

The out-degrees are
deg+(a) = 4, deg+(b) = 1, deg+(c) = 2, deg+(d) = 2, deg+(e) = 3, and deg+(f) = 0.

Source and sink vertex:

In a digraph G, a vertex v with zero in-degree is called a source vertex and

A vertex v with zero out-degree is called sink vertex.

Properties of Relations on Directed graphs:

Let G be a directed graph with vertex set V(G) and edge set E(G) then the digraph G is said to satisfy the following properties.

a) Reflexive: If there is an edge(Vi, Vi) for all Vi belonging to V(G).

b) Symmetric: If whenever the directed edge(Vi, Vj) exists in G, and (Vj, Vi) exists in G.

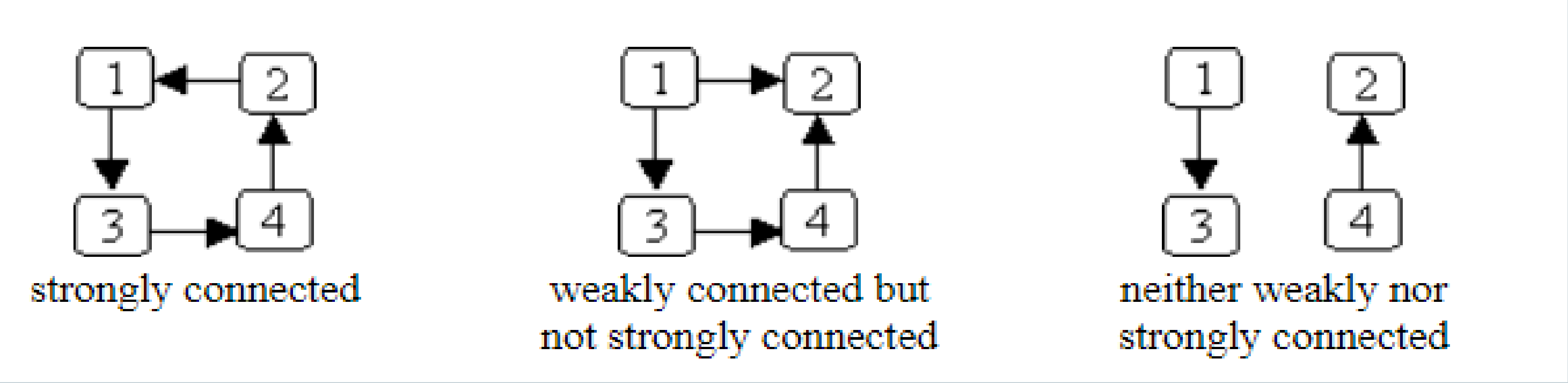
c) Transitive: If whenever (Vi, Vj) and (Vj, Vk) exists in G then (Vi, Vk) exists in G.

Strongly connected:

A ***directed*** graph is **strongly connected** if there is a path from every node to every other node. i.e. any pair of vertices Vi and Vj in G there is a directed path from Vi to Vj and viceversa.

Weakly connected:

A ***directed*** graph is **weakly connected** if, treating all edges as being ***undirected***, there is a path from every node to every other node.



Theorem: The sum of the out-degree of vertices, the sum of in-degree of vertices and The number of edges in a directed graph are equal to each other.

Let $G = (V, E)$ be a graph with directed edges. Then

$$\sum_{v \in V} \deg^{-}(v) = \sum_{v \in V} \deg^{+}(v) = |E|.$$

Representation of Digraph

Adjacency Matrix:

Let G=(v, e) be a digraph with vertices V1, V2,…..Vn. Then the adjacency matrix of G w.r.t given order of vertices is n*n matrix $A(G)= (a_{i,j})_{n*n}$ is defined as follows:

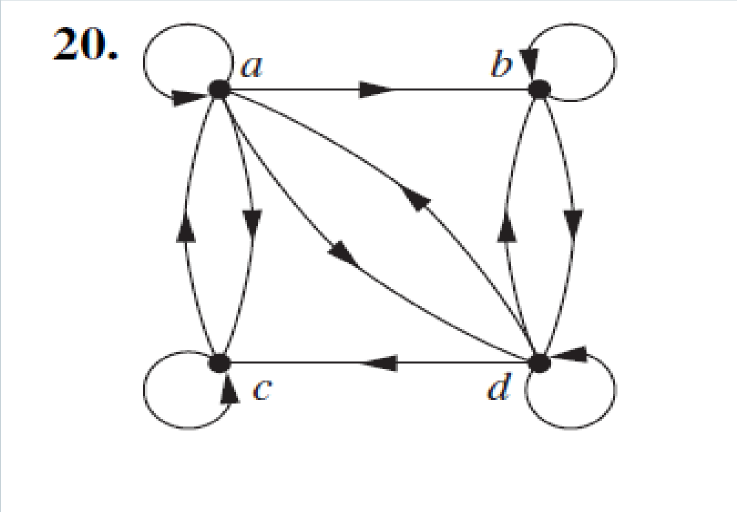
$$a_{i,j} = \begin{cases} 0 & \text{if there is no edge (Vi, Vj)} \\ 1 & \text{if there is an edge (Vi, Vj).} \\ K & \text{if there are K number of edges from Vi to Vj} \end{cases}$$

Incidence Matrix:

Let G be a digraph with vertices V1, V2,….. Vm and directed edges e1, e2,…….en then incidence matrix of G is $I(G) = (m_{i,j})_{m*n}$ is defined as follows:

$$m_{i,j} = \begin{cases} 1 & \text{if edge } e_j \text{ is directed away from a vertex } v_i \\ -1 & \text{if edge } e_j \text{ is directed toward vertex } v_i \\ 0 & \text{otherwise} \end{cases}$$

Find the Adjacency matrix and incidence matrix.



$$\begin{array}{c|cccc} & A & B & C & D \\ \hline A & 1 & 1 & 1 & 1 \\ B & 0 & 1 & 0 & 1 \\ C & 1 & 0 & 1 & 0 \\ D & 1 & 1 & 1 & 1 \end{array}$$

I(G) =

	e1	e2	e3	e4	e5	e6	e7	e8	e9	e10	e11	e12
A	1	0	0	1	0	0	0					
B	1	1	0	0	1	1	2					
C	0	0	1	1	1	0	0					
D	0	1	1	0	0	1	0					